

Education

University of Illinois Urbana-Champaign (UIUC)

Aug. 2026 – Jun. 2028

M.Eng. in Electrical and Computer Engineering

China Jiliang University

Sep. 2022 – Jun. 2026

B.Eng. (Honors) in Optoelectronic Information Science and Engineering

GPA: 93/100 | Rank: 1/223

Internship

- **ByteDance | Data-aml | AI Automobile LLM/VLM Algorithm Intern** Feb. 2026 – Jul. 2026
 - Built an image captioning system for autonomous driving semantic understanding and cockpit HMI; vectorized images with an embedding model, selected cluster-center samples via **K-Means** for **SFT cold start**, and designed a two-stage **RLVR + GRPO** pipeline using JSON field-level TP/FN/FP matching and decoupled LLM-based VQA accuracy.
 - Fine-tuned doubao-embedding-vision and seed-series VLMs for autonomous driving road element recognition; applied **FullSFT** and **LoRA** with positive/negative ratio ablations, **hard negative** construction, and model-forgetting detection; improved average Precision@K from 0.196 to **0.839**, achieving **59.73%** Micro-F1 and **54.67%** Macro-F1.
 - Built an end-cloud offline LLM for always-on, wake-word-free intelligent cockpit; trained a high-precision, low-recall local rejection and **FunctionCall** parser with **SFT + ToolRL**, complemented by cloud-side high-recall rejection, two-stage LLM fallback, and Advisor arbitration; applied **strict/normal dual-mode** post-processing and **dynamic semantic library injection**, achieving **<100ms** latency, **≥95%** instruction accuracy, and **≥98%** rejection accuracy.
- **XPeng Motors | Intelligent Cockpit Center | VLM Foundation Model Intern** Oct. 2025 – Jan. 2026
 - Optimized an image captioning model for higher information density and fewer hallucinations; applied **GRPO** with **decoupled VQA rewards** and **RLVR** to maximize downstream VQA accuracy; built a dual-verification data pipeline to filter information leakage, enforce vision-dependent answering, and improve training data **signal-to-noise**.
 - Improved VLM spatial relationship understanding and reasoning; built spatial benchmarks and aligned open-source model metrics, then applied **SFT cold start** and **GRPO** with custom spatial-reasoning rewards on curated data; achieved **SOTA** on CountBench and RefSpatial Bench.
- **Trusight Technology | Trusight AI Research | Computer Vision R&D Intern** Jul. 2025 – Sep. 2025
 - Optimized object tracking robustness under occlusion, fade-in/out, and tracking-loss scenarios; led the transition from Tracking-by-Detection → Tracking-by-Template → **Point tracking**; benchmarked, optimized, deployed CoTrack, TAP-Next, Locotrack in real-world scenarios; improved tracking accuracy by **10%+** and computational **efficiency** by **2x**.

Research Experience

- **Research Assistant | PLAN Lab, University of Illinois Urbana-Champaign** Jul. 2026 – Present
 - **Latent Embodied Agent:** Extending 3D VLMs with continuous latent reasoning to integrate 3D perception, visual grounding, and spatial reasoning for spatially consistent and computationally efficient multi-step planning.
 - **Latent Reasoning:** Designing anchor-guided bidirectional latent reasoning with sparse spatial anchors and adaptive refinement for accurate and robust 2D/3D spatial reasoning.
- **3DMM-FLUX: Instruction-Guided Facial Expression Editing | Undergraduate Thesis** Mar. 2026 – May 2026
 - [🔗 Code] [🔗 Dataset] [🤗 Hugging Face]
 - Proposed an instruction-driven expression editing framework to address **identity drift** and **fine-grained control** in diffusion-based editing, using **FLUX** as the generative backbone and **DECA**, **FLAME**, **Nvdiffrast** for differentiable 3DMM control; designed a **Conditional DECA Encoder** and **RCG** inference strategy, achieving **AU Acc↑22.2pp to 81.4%**, **ID Sim↑0.072 to 0.824**, and **FID↓8.7 to 15.6**.
- **Research Assistant | Institute of Software, Chinese Academy of Sciences** Sep. 2025 – Oct. 2025
 - Analyzed AI Scientist, an automated scientific paper generation system, covering idea generation, novelty retrieval, automated experiments, paper writing, and review; clarified prompt structures and stage-wise interaction patterns.
 - **AIGC Text Detection:** Built a **RoBERTa + AdaLoc** sentence-level detector with **sliding-window voting** and **overlapping-window aggregation**, achieved **92%** sentence-level accuracy in distinguishing human from AI.
- **Text Vectorization for Illegal Advertising Recognition** Aug. 2024 – Sep. 2024
 - Compared the recognition efficiency of **One-Hot**, **TF-IDF**, and **Word2Vec** text vectorization methods in long-text illegal advertising classification; contributed to 1 EI conference [[Paper](#)] at NLPPIR 2024 and 1 [[software copyright](#)].

Competitions

- (* solo or team leader for all entries)
- **Kaggle: Student Math Misconception Analysis | Silver Medal** Sep. 2025
 - Built a **MAP@3**-optimized misconception classifier for math problem responses; applied feature engineering, **classification-as-generation** reformulation, SFT, and weighted ensembling to fuse multi-model predictions; deployed GPTQ 4-bit quantized with vLLM for low-latency, high-throughput serving, achieving test-set score **↑20% to 97.83%**.
 - **China International College Students' Innovation Competition | Prov. Silver** Aug. 2025
 - Participated in the R&D of a power-compensation battery isothermal calorimeter, responsible for system construction and data processing. Authorized 3 [**software copyrights**], 1 invention [**patent**], and 1 utility model [**patent**].
 - **National Mathematical Modeling Competition | Prov. First Prize** Nov. 2024
 - Led the team to complete a crop planting strategy optimization problem, responsible for data preprocessing, building a multi-objective optimization model, and solving for the optimal planting plan.
 - **National Physics Experiment & Technology Innovation Competition | Prov. Silver** Sep. 2024

Honors & Awards

- **Undergraduate Honors:** CJLU Outstanding Graduate, Honors Graduate of Liangxin Honors College, Merit Student (3×)
- **Scholarships:** 2025 Zhejiang Provincial Government Scholarship, 2026 Guangzhou High-Speed Railway Metrology Scholarship, First-Class School Scholarship (4×)

Skills

- **Languages:** Chinese (Native), English (Proficient), IELTS 7.0 (L6.5, R9.0, W7.0, S6.0), GRE 327 (V157, Q170, W3.5).
- **Programming:** Python, Matlab, C/C++
- **OS:** Linux, MacOS, Windows
- **Frameworks & Libraries:** Pytorch, TensorFlow, JAX, OpenCV, Swift, vLLM, DeepSpeed, NumPy, Pandas, Scikit-learn
- **ML/DL:** LLMs/VLMs Post-training, RL, RLVR, AIGC, Agentic RL, Quantization, Object Det/Trk/Seg.